Supporting Information: Teaching adsorption chemistry by constructing surface complexation models (SCM) in PHREEQC (student handout)

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Adsorption modelling with PHREEQC (SCM workshop): Student Handout

1. Aims and Objectives

- We will look at how comprehensive characterisation of groundwater physicochemical properties (e.g. pH, dissolved oxygen, competitor ions, mineralogy etc) can be implemented in a Surface Complexation Model (SCM) that is sensitive to real-life environmental conditions.
- We will learn how to use the freely-available software package PHREEQC to explore the geochemistry of arsenic and fluoride contamination in the aquifers of central Mexico, looking at saturation indices of various minerals as a function of pH, referencing geothermal waters [6].
- We will learn how Surface Complexation Modelling can be implemented within PHREEQC to provide a computational model of adsorption that allows us to numerically understand how changes in environmental water parameters influences the performance of water treatments based upon adsorption [7].

In this exercise, we will become familiar with geochemical modelling software, where we define chemical problems in terms of equilibrium constants and probe the sensitivity of our system to changes in various water quality parameters.

You will use the geochemical modelling software 'PHREEQC' to develop a surface complexation model, which you will use to predict adsorption of arsenic over goethite mineral surfaces. The practical exercise aims to enable us to:

1. Understand the key concepts in geochemical modelling
2. Set up a geochemical model by defining solutions, solid surfaces, and reaction equilibria
3. Perform modelling experiments to determine how changes in pH and ionic strength can affect adsorption and mobility of toxic metals such as arsenic, in the groundwater system.

Relevant literature:

1. Comparison of Arsenic(V) and Arsenic(III) Sorption onto Iron Oxide Minerals: Implications for Arsenic Mobility by Dixit and Hering, 2003 [1]
   https://pubs.acs.org/doi/10.1021/es030309t
2. Example 8 (Surface Complexation Model) in the PHREEQC user manual (the manual both comes with the software and can be found online) [2]
   https://wwwbrr.cr.usgs.gov/projects/GWC_coupled/phreeqc/phreeqc3-html/phreeqc3-63.htm

2. Introduction

2.1. Limitations of adsorption isotherms

Many important geochemical reactions occur at the interface between fluid and solid phases. One important reaction is the adsorption and desorption of heavy metals onto minerals, which controls the mobility of these toxic elements in groundwater. Adsorption to a solid surface can occur in a number of ways, principally (a) physisorption – a physical bonding between sorbent (solid) and sorbate (the adsorbed molecule), due to van der Waals interactions, or (b) chemisorption – formation of a chemical bond (ionic or covalent) between sorbent and sorbate. When chemisorption occurs, we produce a 'surface complex' – which is analogous to a metal-ligand complex in solution. When the sorbate retains its solvation sphere (water ligands) and only binds electrostatically, we term this an 'outer sphere complex.' When a new covalent bond is formed directly between the sorbate molecule and solid surface, we term this an 'inner sphere complex.'

We most commonly model adsorption processes using adsorption isotherms. These are generally (a) the Langmuir model for monolayer adsorption (the solid surface is covered with a single layer of sorbate molecules), or (b) the Freundlich model for non-monolayer adsorption. (c) The BET adsorption isotherm is used for more complex processes, such as monolayer physisorption followed by filling of pores. These models have very few parameters – essentially defining how much sorbate binds at monolayer coverage, and how strong the sorbate-sorbent interaction is. As such, each isotherm model is only valid under the precise conditions under which the adsorption experiment was carried out – these models have no sensitivity to changes in pH, ionic strength, addition of competitor ions, and all the other factors that geochemists need to consider in natural systems.

2.2. Surface Complexation Models

Instead, to model the influence of these environmental conditions on adsorption, we turn to Surface Complexation Modelling (SCM). This is a more thorough chemical description of the system. We consider sorbate binding to specific sites at the surface (normally surface hydroxyls, S-O-H), and model adsorption as equilibria processes. Each surface complexation reaction is described by a chemical equilibrium equation with an equilibrium constant, log(K). By incorporating log(K) for different species in our system (for instance we will always have carbonate in our system, which weakly binds to all kinds of surfaces), we can already include sensitivity to competitor species.

Many sorbates (e.g. HCO_3^-, $HAsO_3^{-2}$, Na^+, Cl^-, Pb^{+2} etc) are charged, and this will affect how strongly surface complexation occurs. If the mineral surface is positively charged, positive sorbate ions will be repelled and so bind less strongly, whilst negative sorbates will bind more strongly. The surface charge of minerals is due to the acidity of surface hydroxyl groups (S-O-H, where S stands for 'solid'), which can be deprotonated at high pH ($SOH + {}^-OH = SO^- + H_2O$) or protonated at low pH ($SOH + H^+ = SOH_2^+$). Surface charge can also arise through adsorption of electrolyte ions. It is therefore very common to incorporate surface charge into the Surface Complexation Model (SCM). The potential energy is strongest at the surface, and diminishes as we move towards solution, as the surface charge becomes countered by a build-up of electrolyte counter-ions close to the solid surface. This is called the 'Electrical Double Layer,' as we have two layers of charge – (a) the plane of the solid surface, and

(b) the region of counter ions with opposite charge to balance the charge to neutral. There are a number of ways to model the EDL, PHREEQC has both the Diffuse Double Layer model (DDL) and CD-MUSIC (Charge Distribution MUlti-Site Complexation). We will use the simpler DDL model.

2.3. Geochemical modelling with PHREEQC

PHREEQC is one example of a computer program developed for solving geochemical problems. PHREEQC has 'master species' and 'species.' Each species is a type of chemical or a molecule, and the master species are the building blocks that can be used to build each molecule. If we could consider the 'master species' CO_3^{2-} and H^+, then we can combine these blocks in four ways, to create four different 'species'

- H^+
- CO_3^{2-}
- HCO_3^-
- H_2CO_3
- (of course, we can also have the species ^-OH by subtracting 1 H^+ from nothing)

By defining the equilibrium constants for the formation of each species from its master species, we can generate our chemical speciation problem. Modelling this in PHREEQC we could easily calculate the changes in carbonate speciation as a function of pH or whatever other water quality parameters we are interested in. PHREEQC also incorporates solids and precipitation/dissolution reactions, as well as solid surfaces, which we are interested in. Here we have 'surface species' which are our surface complexes, and are built from combinations of our 'surface master species' and the solution species. Example 8 in the PHREEQC 'Help' files is a good example of a surface complexation model

PHREEQC can be a very powerful tool as it already incorporates a number of databases – these are libraries where plenty of common environmental reactions are already defined. PHREEQC also incorporates the Visual Basic programming language, so we can rapidly generate large data sets, looking at changing multiple variables at the same time.

3. Exercises

In this exercise, we will use SCM parameters (both equilibrium constants and adsorbent physical properties) from the literature: the adsorption of As(V) onto goethite from the paper *Comparison of arsenic(V) and arsenic(III) sorption onto iron oxide minerals: Implications for arsenic mobility* by Dixit and Hering. In this paper, you will find data tables containing all necessary information. For each exercise, save your input and output files with a unique name, so that you can backtrack if an error occurs.

3.1. Creating a simple aqueous system

1. Make sure you have PHREEQC on your computer. PHREEQC can be obtained via the following
    a. Download PHREEQC from the United States Geological Survey https://wwwbrr.cr.usgs.gov/projects/GWC_coupled/phreeqc/
        i. Use the .msi Microsoft installer
2. Many geochemical software programs contain libraries or databases filled with all kinds of reactions ready for us to use. In PHREEQC, the most comprehensive database is llnl.dat. Change the PHREEQC database to llnl.dat
    a. Options → Set Default Database then click "…" to browse
3. First consider just the aqueous environment (no goethite, no surface complexation). You will have to define your solution master species
    a. Add the keyword block SOLUTION_MASTER_SPECIES
    b. On the next line define the As(V) master species by entering
       As(+5) H2AsO4- 0.0 As 74.92
    c. Finish the block with a new line stating END
4. Next define the different solution species which can be formed by the sequential protonation of the arsenate anion
    a. Start the keyword block by typing SOLUTION_SPECIES
    b. Type in the reaction formula for the first protonation
       AsO4-3 + H+ = HAsO4-2
    c. On the line underneath define the equilibrium constant for this reaction by typing
       log_k 11.60
    d. On the next lines add the other possible species by writing reactions for the arsenate anion plus two and plus three protons, using the equilibrium constants from the literature
    e. END this keyword block
5. Now you have defined the possible chemical species and the equilibrium constants which govern the balance between them, you need to define the aqueous system and it's conditions
    a. Create a new solution with the ID of "1" by typing SOLUTION 1
    b. Define the pH as 5 as follows
       pH 7.0
    c. Add 0.1 M of chloride by typing
       Cl 0.1 mol/kgw
       (where mol/kgw stands for moles per kilogram weight, equivalent to mol L^{-1})

   d. Add the same amount of sodium (it's important to ensure the charges balance by having equivalent concentrations of cation and anion)
   e. Add 100 ppb (ug L-1) arsenate as follows
      As(+5) 100 ug/kgw
   f. END the solution block
      i. END tells PHREEQC that we are ready to run a simulation. We can have END several times in a single input file, and each time END is reached the software will perform the necessary calculations.
6. Now run your first model
   a. Click RUN and Start
   b. You will see an Output file opened (and also saved to your folder)
   c. Which is the most prominent arsenic species under these conditions? What are the concentrations of each species?
7. Try changing the pH from 7 to 3 and then to 11, running the model again each time.
   a. How does the speciation of arsenic change with pH? What do we see, and how can we explain the pH dependence of arsenic speciation?

3.2. Investigating the pH dependence of Arsenic speciation

1. Now let's calculate the speciation of arsenic at all pH values between 3 and 11
2. Changing the pH, hitting RUN, checking the result, changing the pH again is slow and tedious
3. You will find it quicker to create new solutions. Simply copy and paste your solution block as many times as needed, changing the pH to the desired value (e.g. 3.0, 4.0, 5.0 etc)
   a. Remember to update the SOLUTION ID
4. To collect all the data together, let's print the output as an Excel spreadsheet
   a. Before the SOLUTION keyword blocks, add in the SELECTED_OUTPUT keyword block
   b. Define the output file name as follows
      -file output.xls
   c. Then add
      -reset true
   d. Then we need to define which species we want to know the concentration of. We do this as follows
      -molalities AsO4-3 HAsO4-2 …
      (add the formulae of all species we need – there are four)
   e. END the keyword block before the start of the SOLUTION blocks
5. RUN the model
6. Do you see the new output file? Open it in Excel. You should have a number of rows, one for each solution you have simulated, each with a unique pH value. On the right hand side you will see the concentrations (mol L-1) of each of the arsenic species you request with the –molalities variable.
7. Make a scatter plot of your results
   a. Click on the spreadsheet somewhere away from the data
   b. Click INSERT → Scatter plot (use smoothed lines without markers → this is more appropriate for modelled data
   c. Right click on the empty graph → SELECT DATA
   d. Enter the name for the first species, e.g. AsO4-3

   e. For the x-values drag around the pH data (column G for me)
   f. For the y-values select the data under the column labelled "m_ASO4-3"
   g. Repeat for each species
   h. Add axis titles and a legend to your figure
   i. Save the spreadsheet with a new name so that we don't accidentally overwrite it
8. You will have to close the output spreadsheet each time you want to run the model again (PHREEQC won't overwrite the output file)
1. **Optional:** Next, we will add a calculation of saturation indices to ensure that we are running our experiment at conditions in which As(V) does not precipitate out of solution. This is a useful check to ensure that any reduction in the concentration of your contaminant after water treatment is due to surface adsorption, and not due to bulk precipitation.
   a. We will ask PHREEQC to provide the saturation index of As_2O_5 in the Excel file output. To do this, at the end of your SELECTED_OUTPUT keyword block, before the END statement, type
      -saturation_indices As2O5
   b. Run the simulation and you're done! We know that when the saturation index is greater than zero, the solution is saturated, and we expect bulk precipitation to occur. (We can model bulk precipitation using EQUILIBRIUM_PHASES, for which there are many examples available online, in the user guide). Are our simulations saturated and would we expect the precipitation of As_2O_5? How does the saturation index change with pH? Does this trend make sense based on the chemistry?

3.3. The surface acidity of goethite

1. Next, we want to introduce the goethite mineral to the model. We will first look at the surface acidity – the protonation and deprotonation of S-OH surface hydroxyl groups. These are the active sites where arsenic will bind.
2. Below your SOLUTION_SPECIES keyword block, start the key word block SURFACE_MASTER_SPECIES
   a. Below this define the surface master species S-OH by typing
      Surf_ Surf_OH
      This is the component we will use to build our surface complexes from
3. Next start the SURFACE_SPECIES keyword block
   a. Considering surface acidity, we have three possibilities for the surface hydroxyl
      i. Remaining unchanged as S-OH (or Surf_OH)
      ii. Becoming protonated as $S\text{-}OH_2^+$ (or Surf_OH+2)
      iii. Becoming deprotonated as $S\text{-}O^-$ (or Surf_O-)
   b. Type in the formulae for each reaction, much like you did for the aqueous reactions. Similarly add the log_k values, using the acidity constants defined in the literature
      i. Note that we need to define the species S-OH as the component S-OH, with a log(K) value of zero. So one of your equations will have to read
         Surf_OH = Surf_OH
         log_k 0.0
4. Now we have our surface acidity reactions, we need to create the surface, just like we previously created our solutions

   a. Start the keyword block
      SURFACE 1
   b. State that we will define the concentration of surface hydroxyls in terms of 'site density' which takes the units 'sites nm^{-2}', by typing
      -sites DENSITY
      i. We could alternatively define our surface hydroxyls as 'mol L^{-1}'
   c. Define the surface by typing
      Surf_OH <site density> <surface area> <concentration>
      where site density (sites nm^{-2}), surface area (m^2 g^{-1}) and solid concentration (g L^{-1}) are all found in the literature
   d. END the block
5. What we do next is mix the surface with the solution and see what speciation we get once we reach equilibrium. We only need one solution to do this, so you can delete all solutions except for SOLUTION 1, which should be at pH 1
   a. Change the arsenic concentration to zero, since we will just look at surface acidity
6. We can use the PHASES keywords to mix the surface and solution, buffering to a specific pH
   a. Underneath the END of SOLUTION 1 add the following
      PHASES
      Fix_H+
      H+ = H+
      log k 0.0
      END
7. Then we will mix the solution and the surface as follows
   USE SURFACE 1
   USE SOLUTION 1
   EQUILIBRIUM_PHASES 1
   Fix_H+ -3.0 NaOH 11.0
   END
   a. This is a 'pseudo-phase' that adds sodium hydroxide until we reach the pH needed
      i. Here we want pH 3 so we set Fix_H+ to -3.0 (minus pH…)
8. Copy and paste the input data from step 7, and now change the -3.0 to -3.5
   a. This will increase the pH to 3.5
   b. Repeat this until we have the whole pH range from 3 to 11 in 0.5 step intervals
9. Change the SELECTED_OUTPUT –molalities to the species we want to look at (Surf_OH Surf_OH2+ Surf_O-)
10. Run the simulation
11. Add a new column and call it Q (which stands for surface charge). To determine Q we can simply take the concentration of SOH_2^+ and subtract SO^-. Do this.
12. Plot Q, [SOH], [SOH_2^+] and [SO^-] as a function of pH
13. How does surface charge change as a function of pH? Can you explain this?

3.4. Surface complexation of As(V) onto goethite

1. We have now successfully calculated (1) the arsenic speciation in aqueous solutions, and (2) the surface charge of goethite, both as a function of pH.
2. Next, we will add in the arsenic surface complexation reactions to complete our model

a. In the SURFACE_SPECIES keyword block, define your possible surface complexation reactions between S-OH and arsenic, adding in the equilibrium constants from the literature
    i. Your first entry will be
    Surf_OH + AsO4-3 + 3H+ = Surf_H2AsO4 + H2O
    log_k 31.00
    ii. Add all three possible complexes
b. Set the concentration of arsenic to 100 uM, just like in the literature paper
    i. You could also change the ionic strength from 0.1 to 0.01 M if you want to match the paper more closely
c. Change the species to be printed into the output file, by going to the SELECTED_OUTPUT keyword block and changing the –molalities
    i. You will probably want to print each of the different aqueous arsenic species, and each of the arsenic surface complexes
d. Run the model, and in your output spreadsheet add a new column, labelling it as 'Total adsorbed Arsenic'
    i. This column needs to be the sum of the concentrations of each arsenic surface complex
e. Plot the total adsorbed arsenic, as well as each individual surface complex species, as a function of pH
f. What is the trend in total adsorption with pH? How can you explain this? (Think about the surface charge of goethite as a function of pH, and the charge of the aqueous arsenic species)
g. At which pH do we see each individual surface complex appear? Can you explain this?

3.5. Experiments

1. Sea water is approximately 0.7 M NaCl. What happens to our surface complexation model as we move from 0.1 M NaCl to 0.7 M NaCl?
    a. Run the model at a range of different ionic strengths, collecting all the results and plotting together
    b. Describe what we see and explain this where possible
2. Next let's look at a 'breakthrough' curve. Say we make a filter system to remove arsenic from drinking water by adsorbing it to goethite, we need to understand how much arsenic we can remove before the material is saturated and needs to be replaced.
    a. Change the concentration of goethite to 10 g L^{-1}
    b. Change the ionic strength to 0.01 M
    c. We will now need to copy and paste the solution keyword block, changing the amount of arsenic in each solution
    d. You can use the following initial concentrations:
    0, 0.1, 1, 10, 20, 30, 50, 75, 100, 150, 200, 300, 400 and 600 ppm (mg/kgw) As(V)
    e. Now you will need to 'USE' your SURFACE 1 and a solution, then run EQUILIBRIUM_PHASES, with the pH fixed to a specific value (I used pH 7)
        i. So change the number of solution in each of your equilibrium phases blocks
    f. Run the model, and there are two ways of plotting the data
        i. As(V) (aq) as a function of total As(V) added (this is a breakthrough curve)

   ii. As(V) (adsorbed) (mg arsenic per gram sorbent) versus As(V) (aq) (this is an 'adsorption isotherm'
   iii. Try plotting the data in both ways, you will have to normalise your data to one gram per solid (you are using 10 g L^{-1})
3. Finally, try changing the ionic strength of your solutions to investigate the effect of salinity.
   a. Simply change the concentration of Na and Cl and run the model again. Each time your run the model, add the output data as a new curve to your graph.
   b. How does salinity affect the breakthrough curve and adsorption isotherm? Is this good or bad for arsenic remediation (cleaning up drinking water)?
4. Calculate the pH adsorption-edge for As(III). How does this compare with the As(V) adsorption edge? Can we explain this in terms of surface charge and the charge of the aqueous arsenic species
   a. Note that the llnl.dat database already has aqueous speciation for As(III) (and As(V)). The As(III) species within llnl.dat contain aqueous species we do not want in our simulations: $HAsO_2$ and $AsO2^-$ (where the arsenic atom is erroneously coordinated to two oxygen atoms only) and AsO_2OH^- and $As(OH)_3$ (which both clash with the H_3AsO_3 notation used throughout this practical). You will want to disable these species by editing the llnl.dat (which can be found in the Program Files folder containing the PHREEQC installation) and deleting the reactions or commenting them out using the '#' symbol (you can find each reaction by using Find, or Ctrl + F). A modified version of llnl.dat with these corrections is also available within the Supporting Information.
   b. After running the simulation, note the presence of As(V) in the output file. Some of the As(III) has been oxidised. How do we explain this? Search online for a Pourbaix diagram of aqueous arsenic to understand the relationship between pH, redox potential, and oxidation states.

3.6. Coding loops using Visual Basic for more advanced simulations

1. A key advantage to using computational tools should be the ability to simulate large data sets. Up till now, we've been manually copying, pasting and editing key word blocks in able to run simulations with different pH values and different adsorbate concentrations. Some of the simulated curves were not so smooth when plotted in Microsoft Excel due to the limited resolution in the data set. Ideally, we would fill in the gaps between these data points with additional simulations!
2. The most effective way to do this, is to code in a loop. PHREEQC allows us to do this easily, integrating the Visual Basic programming language.
3. Open up the input file from your pH adsorption edge exercise and save it with a new name.
4. You should have about 17 blocks similar to
```
USE SURFACE 1
USE SOLUTION 1
EQUILIBRIUM_PHASES 1
Fix_H+ -3.0 NaOH 11.0
END
```
5. Delete these repeating blocks
6. We can replace this with a loop that creates a text file (.txt), writes in potentially thousands of simulations, and then reads it back into PHREEQC. To do this, go to your SOLUTION 1 key word block and create a new line above the END statement (the Visual Basic code has to go within a keyword block).

7. Enter the following code:
```
SELECTED_OUTPUT
     -file 07_loops_example_USER_PUNCH.txt
     -reset false
USER_PUNCH
10 FOR j = 3 to 11 STEP 0.1
20 a$ = a$ + EOL$ + "USE surface 1" + EOL$
30 a$ = a$ + EOL$ + "USE solution 1" + EOL$
40 a$ = a$ + "EQUILIBRIUM_PHASES 1" + EOL$
50 a$ = a$ + "   Fix_H+ " + STR$(-j) + " NaOH 10.0" + EOL$
60 a$ = a$ + "END" + EOL$
70 NEXT j
80 PUNCH a$
```
8. Try to understand the code.
   a. SELECTED_OUTPUT tells PHREEQC that we wish to print some kind of output
   b. -file indicates the name of the file we wish to save to
   c. USER_PUNCH indicates that we are going to manually tell PHREEQC what we would like the output to be
   d. 10, 20, 30, 40 etc… are all numbers used to indicate which lines of the Visual Basic code come first. You can also number the lines 1, 2, 3, 4 or 10, 25, 64, 88 … etc, as long as the numbers increase chronologically.
   e. FOR j = 3 to 11 STEP 0.1. Here we create a variable called *j*. This variable starts at 3 and increases to 11 in 0.1 increments. So we cycle through j = 3, 3.1, 3.2, 3.3, 3.4 … 10.7, 10.8, 10.9, 11.0. These are the pH values we wish to simulate
   f. a$ is a string variable
   g. EOL$ is a special character to create a line break. It is like hitting the return key.
   h. `20 a$ = a$ + EOL$ + "USE surface 1" + EOL$` will add a new line to our 07_loops_example_USER_PUNCH.txt text file that simply says "USE surface 1".
   i. `50 a$ = a$ + "   Fix_H+ " + STR$(-j) + " NaOH 10.0" + EOL$` will create a new line that says "Fix_H+ -3.0 NaOH 10.0" or "Fix_H+ -3.1 NaOH 10.0" or whatever our pH value is.
   j. NEXT j tells the FOR loop to repeat to the next value of j (our pH value)
   k. PUNCH a$ tells PHREEQC to write the entire contents of a$ (all our lines of code) to the text file 07_loops_example_USER_PUNCH.txt.
9. Your END statement for the SOLUTION 1 keyword block should still come after the Visual Basic code. Your input file should then have the PHASES keyword block, and the SELECTED_OUTPUT keyword block which tells PHREEQC to save the concentrations of each species to output.xls
10. Finally, above the final END statement (in the second SELECTED_OUTPUT keyword block), add a couple of lines. Add the following code:
```
USER_PUNCH
 10
INCLUDE$ 07_loops_example_USER_PUNCH.txt
```
11. Here, you tell PHREEQC that you'd like to enter some code again. INCLUDE$ opens up the .txt file that your script previously wrote all the pH adjustment code to.
12. Click run. You should see that a new ,txt file is created containing the input for all your simulations, with the pH varying between 3 and 11 in 0.1 pH unit increments. The output.xls file should contain about 80 rows of data. Plot the results. Well done!

4. References

[1] S. Dixit, J. Hering, Comparison of arsenic(V) and arsenic(III) sorption onto iron oxide minerals: Implications for arsenic mobility, Environ. Sci. Technol. 37 (2003) 4182–4189. https://doi.org/10.1021/es030309t.

[2] PHREEQC Version 3, (n.d.). https://www.usgs.gov/software/phreeqc-version-3 (accessed June 7, 2020).